



- 4 (a) State what is meant by simple harmonic motion.

.....

.....

..... [2]

- (b) A block is suspended from a spring, as shown in Fig. 4.1.

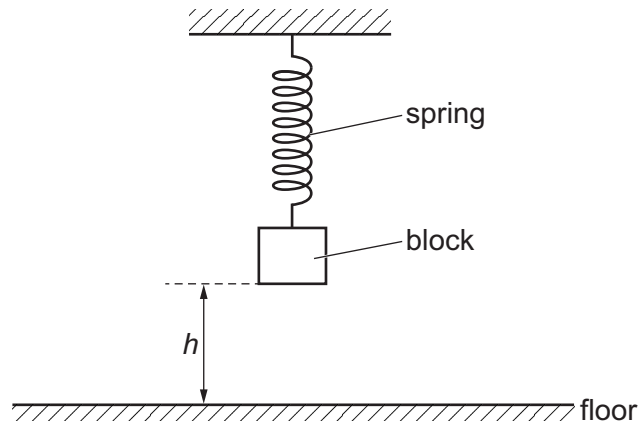


Fig. 4.1

The block is pulled down and released at time $t = 0$. It then oscillates vertically with simple harmonic motion.

Fig. 4.2 shows the variation of the velocity v of the block with height h of the base of the block above the floor.

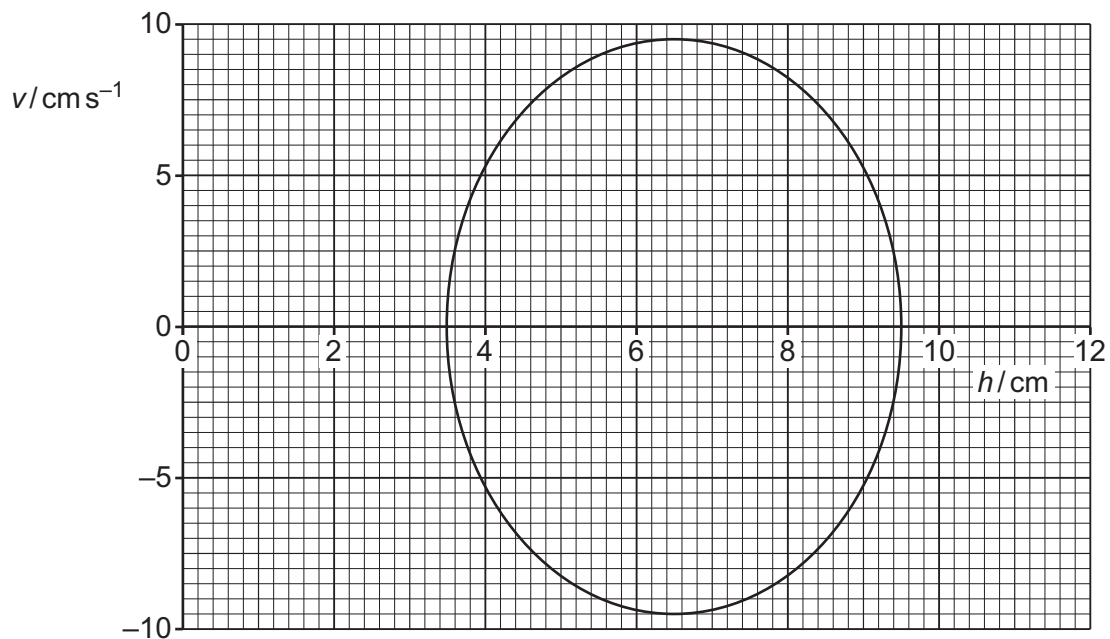


Fig. 4.2



- (i) Determine the amplitude, in cm, of the oscillations.

amplitude = cm [1]

- (ii) Show that the angular frequency of the oscillations is 3.2 rad s^{-1} .

[2]

- (iii) Calculate the period T of the oscillations.

$T = \dots\dots\dots$ s [2]

- (iv) On Fig. 4.3, sketch the variation of h with time t from $t = 0$ to $t = 6.0$ s.

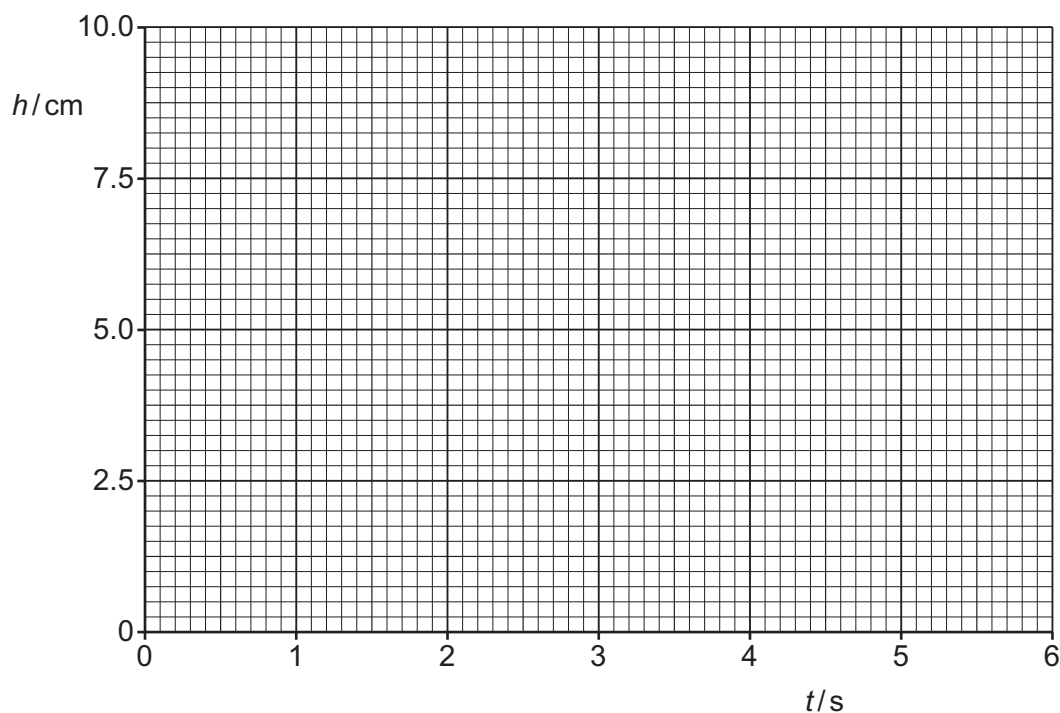


Fig. 4.3

[4]

[Total: 11]